

Yixin Chen

PhD candidate, Peking University

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RESEARCH PROFILE

I work on medical image segmentation and learning across PET, CT and MRI. My clinical research focuses on cancer, fatty liver disease and multi-organ metabolism.

EDUCATION

Peking University

PhD in Medical Technology

September 2023-present

King's College London

MSc in Artificial Intelligence, with Distinction

2019-2020

Xiamen University

BSc in Automation; second bachelor's degree in Mathematical Finance

2015-2019

CHALLENGE RESULTS AND RESEARCH FUNDING

autoPET V: second place in Category 1

MICCAI autoPET V challenge, interactive PET/CT lesion segmentation.

2026

CMR-MULTI: second place

Multi-sequence cardiac MRI segmentation. Method paper selected for a CMRSeg challenge oral presentation.

2026

REG²: fourth place

Method paper selected for a MICCAI 2026 challenge workshop poster.

2026

NSFC doctoral research project

Principal investigator of a project on extracting and analysing multi-organ metabolic information from total-body PET/CT.

2026-2027, NSFC grant for doctoral students

PROFESSIONAL EXPERIENCE

AirDoc

Algorithm Engineer

Developed a model to predict the gap between an implanted ICL lens and the natural lens after surgery, using eye measurements and lens size.

Worked on ultra-widefield fundus image analysis, clinical requirements and medical-device registration.

July-September 2022

BioMind

Medical Imaging Algorithm Engineer

Developed cardiac ultrasound models for image quality assessment, Doppler segmentation and measurement, septal defect detection and valve regurgitation analysis.

Worked with clinicians to integrate models into an ultrasound-assisted diagnosis system. Also contributed to an ovarian-tumour ultrasound system.

November 2020-June 2022

2026

Whole-Body [¹⁸F]FDG PET/CT Characterizes Multi-Organ Metabolic Alterations in MASLD

EJNMMI, 2026
Accepted

Yixin Chen (first author), Chuanzhi Zhou, Lijuan Guo, et al.

NeuronCtrl: Geometry-Aware Safe Closed-Loop Generative Control for Neuronal Microenvironment Dynamics

ICML, 2026
Spotlight

Haowei Xu, **Yixin Chen**, Wanyi Fu, Hongbin Han, Zhaoheng Xie
Conference OpenReview

Neural Network Approximation Aided Kinetic Parameter Inference of Compartment Models in Dynamic PET

IEEE TRPMS, 2026

Wenrui Shao, **Yixin Chen**, Ourui Fu, Tian Li, Zhi Yang, Nan Li, Hongbin Han, Xiangxi Meng, Zhaoheng Xie
Paper

PHATE-Net: Differentiable Pseudotime Learning for Trustworthy Disease Trajectories in PET

CVPR Findings, 2026

Yixin Chen, Yan Wang, Wenrui Shao, Zhaoheng Xie
Paper

2025

Modality-projection universal model for comprehensive full-body medical imaging segmentation

Nature Communications,
2025

Yixin Chen (first author), Lin Gao, Yajuan Gao, Rui Wang, Jingge Lian, Xiangxi Meng, Yanhua Duan, Leiying Chai, Hongbin Han, Zhaoping Cheng, Zhaoheng Xie
Paper arXiv Code

VP-SFDA: Visual Prompt Source-Free Domain Adaptation for Cross-Modal Medical Image

Health Data Science,
2025

Yixin Chen, Yan Wang, Zhaoheng Xie
Paper

Neural network-aided unsupervised input function estimation for dual-time-window PET Patlak analysis

EJNMMI Physics, 2025

Wenrui Shao, Yarong Zhang, Fen Du, Fangxiao Cheng, **Yixin Chen**, Xiangxi Meng, Ying Liang, Zhaoheng Xie
Paper

A Physiology-Supervised Neural Operator Framework for Kinetic Parameter Estimation in Dynamic PET

IEEE NSS/MIC/RTSD,
2025

W. Shao, **Yixin Chen**, Z. Wang, R. Xu, X. Meng, Z. Xie
Paper

2024

IEEE TMI, 2024

PCNet: Prior Category Network for CT Universal Segmentation Model

Yixin Chen, Yajuan Gao, Lei Zhu, Wenrui Shao, Yanye Lu, Hongbin Han, Zhaoheng Xie

Paper Code

MICCAI, 2024

Poster

LUCIDA: Low-Dose Universal-Tissue CT Image Domain Adaptation for Medical Segmentation

Yixin Chen, Xiangxi Meng, Yan Wang, Shuang Zeng, Xi Liu, Zhaoheng Xie

Paper Code

IEEE TRPMS, 2024

Structure-Enhanced Unsupervised Domain Adaptation for CT Whole-Brain Segmentation

Yixin Chen (first author), Yajun Gao, Lei Zhu, Jianan Li, Yan Wang, Jiakui Hu, Hongbin Han, Yanye Lu, Zhaoheng Xie

Paper Code

IEEE NSS/MIC/RTSD,

2024

Neural Network-Assisted Dual-time-window Acquisition: Modeling Incomplete Input Functions for Patlak Analysis

W. Shao, Y. Zhang, **Yixin Chen**, X. Meng, Y. Liang, Z. Xie

Paper

2023

IEEE TMI, 2023

FVP: Fourier Visual Prompting for Source-Free Unsupervised Domain Adaptation of Medical Image Segmentation

Yan Wang, Jian Cheng, **Yixin Chen**, Shuai Shao, Lanyun Zhu, Zhenzhou Wu, Tao Liu, Haogang Zhu

Paper

IEEE EMBC, 2023

Rethinking Disentanglement in Unsupervised Domain Adaptation for Medical Image Segmentation

Yan Wang, **Yixin Chen** (first author), Yingying Zhang, Haogang Zhu

Paper

Frontiers in

Cardiovascular

Medicine, 2023

Echocardiography-based AI for detection and quantification of atrial septal defect

Xixiang Lin, Feifei Yang, **Yixin Chen** (first author), et al.

Paper

2022

IEEE EMBC, 2022

MSGAN: Multi-Stage Generative Adversarial Networks for Cross-Modality Domain Adaptation

Yan Wang, **Yixin Chen** (first author), Wenjun Wang, Haogang Zhu

Paper

Cancers, 2022

A deep learning model system for diagnosis and management of adnexal masses

Jianan Li, **Yixin Chen** (first author), Minyu Zhang, et al.

Paper

Xixiang Lin, Feifei Yang, **Yixin Chen**, et al.
Paper

Preprints

arXiv, 2026

PreSight: Preoperative Outcome Prediction for Parkinson's Disease via Region-Prior Morphometry and Patient-Specific Weighting

Yand Wang, Chen Zhang, Lanyun Zhu, **Yixin Chen**, Qunbo Wang, Yutong Bai, Jurgen Germann, Yinghong Wen, Shuai Shao
arXiv

SSRN preprint, 2025

Whole-Body Inter-Organ Metabolic Atlas for PET/CT Anomaly Detection

Yixin Chen, Lin Gao, Lei Zhu, Ruoyan Xu, Fangxiao Cheng, Xinyu Liu, Hongbin Han, Zhaoping Cheng, Zhaoheng Xie
Preprint

arXiv preprint, 2023

FP-PET: Large Model, Multiple Loss And Focused Practice

Yixin Chen, Ourui Fu, Wenrui Shao, Zhaoheng Xie
arXiv

arXiv, 2023

SL: Stable learning in source-free domain adaption for medical image segmentation

Yixin Chen, Yan Wang
arXiv

Challenge papers and conference abstracts

MICCAI REG² challenge workshop, 2026
Poster

Diagnosis-Conditioned Retrieval for Structured Pathology Workflows

Yixin Chen
Paper

MICCAI CMRSeg workshop, 2026
Selected for oral presentation

Acquisition-Structured nnU-Net Ensembles for Multi-Sequence Cardiac MRI Segmentation and Quantification

Yixin Chen, Yizhen Peng
Paper

MICCAI MVAA challenge workshop, 2026
Poster

Failure-Mode-Driven Multimodal Mitral Valve Segmentation for the MVAA 2026 Challenge

Yixin Chen
Paper

EANM, 2025
Oral abstract, OP-308

Multiorgan Metabolic Dysregulation in Alzheimer's Disease: Total-body PET/CT Reveals Systemic Network Interactions with Amyloid Burden

R. Xu, Y. Duan, L. Gao, **Yixin Chen**, H. Han, Z. Cheng, Z. Xie
Programme

PATENTS, TEACHING AND ACADEMIC SERVICE

12 invention patents

4 granted; 8 under substantive examination.

Nature Communications

Early Career Co-Reviewer

Computer-vision teaching

Recorded introductory and intermediate courses on machine learning and computer vision.

Deep Learning Theory and Practice: PyTorch Case Studies

2021

Sole author. Published by Tsinghua University Press in 2021.